

Week 1 Final Project Report

UCI Adult Income Prediction

1. Problem Definition

The objective of this project was to develop a supervised machine learning classification system that predicts whether an individual's annual income is greater than \$50K using demographic, educational, occupational, and financial attributes.

This problem is relevant because income classification provides a practical example of binary classification with mixed numerical and categorical data, missing values, class imbalance, and potentially important fairness considerations. The project focused not only on predictive performance but also on reproducibility, model interpretation, threshold selection, and production-ready inference.

The target variable was converted into a binary classification problem:

- $0 \rightarrow \text{income} \leq \50K
- $1 \rightarrow \text{income} > \50K

The positive class represented approximately 23.93% of the observations.

2. Data Preparation

The project used the **UCI Adult Income Dataset**, containing 48,842 observations. The dataset includes variables describing age, workclass, education, marital status, occupation, relationship, race, sex, capital gain, capital loss, working hours, and native country.

The data was divided into development and held-out test sets. The development data was further divided into training and validation sets. Stratified splitting was used to maintain a consistent class distribution across the subsets.

The preprocessing pipeline included:

Numerical features

- Median imputation for missing values
- StandardScaler normalization

Categorical features

- Most-frequent imputation
- One-hot encoding
- `handle_unknown="ignore"` to support unseen categories during inference

Eight engineered features were added:

- age_group
- hours_group
- capital_net
- has_capital_gain
- has_capital_loss
- education_age_ratio
- work_hours_age
- is_married

Feature engineering and preprocessing were integrated into a single scikit-learn pipeline. This ensured that the same transformations were automatically applied during training and inference.

Five-fold stratified cross-validation was used for model comparison.

3. Model Development

Three classification models were evaluated:

- Logistic Regression
- Random Forest
- HistGradientBoostingClassifier

The models were compared using F1 score and ROC AUC because the target variable was imbalanced and accuracy alone would not adequately represent positive-class performance.

Model	Mean F1	ROC AUC
Logistic Regression	0.6693	0.9127
Random Forest	0.6677	0.9024
HistGradientBoosting	0.7063	0.9259

HistGradientBoosting achieved the strongest cross-validation performance and was selected for further tuning.

4. Model Tuning

Hyperparameter tuning was performed on the HistGradientBoostingClassifier to improve predictive performance.

The selected configuration was:

```
learning_rate = 0.0986849497
max_depth = 7
max_iter = 234
```

```
l2_regularization = 8.2443121909
random_state = 42
```

The classification threshold was also optimized using the validation data. Instead of using the default threshold of 0.50, a threshold of **0.35** was selected to improve the F1 score.

The final model was then fitted and evaluated once on the previously held-out test set.

5. Diagnostics

Learning Curve

The learning curve showed that the model initially had a substantial difference between training and validation performance. At 2,735 training examples, training F1 was 0.8768 compared with validation F1 of 0.6563.

As more training data was added, the gap decreased substantially. At 27,350 examples, training F1 was 0.7386 while validation F1 reached 0.7040.

This indicates some initial overfitting on smaller training sets, but the gap narrowed as more data became available. The validation score also continued to improve, suggesting that the final model generalizes reasonably well and that additional training data could potentially provide further improvements.

Calibration

Probability calibration was evaluated using a calibration curve and Brier score.

The final model achieved a Brier score of:

0.08795

The calibration analysis provides an assessment of how closely predicted probabilities correspond to observed positive-class frequencies. The model's probabilities are useful, but calibration is not perfect, which is expected for a tree-based classifier without a dedicated probability-calibration step.

6. Final Results

The final HistGradientBoosting pipeline was evaluated on the held-out test set using the selected threshold of 0.35.

Metric	Test Score
Accuracy	0.8613
Precision	0.6850
Recall	0.7784
F1 Score	0.7287

Metric	Test Score
ROC AUC	0.9280
PR AUC	0.8295

The confusion matrix was:

	Predicted	
	0	1
Actual 0	6594	837
Actual 1	518	1820

The model correctly identified 1,820 positive cases and 6,594 negative cases. It produced 837 false positives and 518 false negatives.

The final F1 score of 0.7287 demonstrates a good balance between precision and recall for the positive income class, while the ROC AUC of 0.9280 indicates strong class-discrimination ability.

7. Model Interpretation

Because HistGradientBoostingClassifier does not provide the standard `feature_importances_` attribute, **permutation importance** was used.

The most influential features were:

Rank	Feature	Importance
1	marital-status	0.1420
2	capital-gain	0.1130
3	education-num	0.0837
4	Age	0.0541
5	Occupation	0.0385
6	capital-loss	0.0276
7	hours-per-week	0.0237
8	Relationship	0.0108
9	Sex	0.0052
10	workclass	0.0038

Education and age generally showed positive relationships with predicted income probability. The model also assigned higher average predicted probabilities to categories such as Married-civ-spouse, Exec-managerial, and Prof-specialty.

These are associations learned from the dataset and should not be interpreted as causal relationships.

The strong influence of demographic variables such as marital status and sex is also important from a fairness perspective. Such variables may encode historical socioeconomic patterns or biases present in the training data.

8. Production Readiness

The complete final machine learning workflow was saved as:

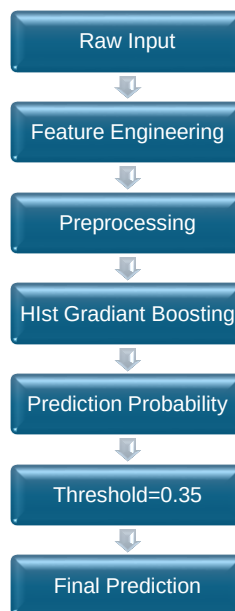
`final_model.joblib`

The selected classification threshold was saved separately as:

`final_threshold.joblib`

The saved model contains feature engineering, preprocessing, and the final HistGradientBoosting classifier. Therefore, new observations can be passed directly to the pipeline without manually repeating preprocessing.

The inference workflow is:



The inference workflow was successfully tested on **10 unseen examples**, producing probabilities and final predictions for all examples.

Reproducibility was supported by controlling model randomness with `random_state=42` and documenting the Python and library versions used.

Environment

Component	Version
Python	3.13.15

Component	Version
NumPy	2.1.3
Pandas	2.2.3
Matplotlib	3.10.0
Seaborn	0.13.2
Scikit-learn	1.6.1
SciPy	1.16.3
Joblib	1.5.3

9. Limitations & Future Improvements

Several limitations remain.

First, the dataset represents historical income patterns and may contain socioeconomic and demographic biases. The use of attributes such as sex, race, and marital status may introduce fairness concerns. Future work should evaluate fairness metrics across demographic groups and consider whether sensitive attributes should be excluded or handled differently.

Second, the model's probability calibration is not perfect. A future version could apply methods such as Platt scaling or isotonic regression using a dedicated calibration set.

Third, additional data could improve generalization. The learning curve suggests that validation performance was still improving as the training set increased.

Feature engineering could also be expanded using domain-informed interactions and more carefully designed nonlinear transformations. Additional models, such as XGBoost, LightGBM, or calibrated ensemble methods, could be compared if available.

Finally, threshold selection could be adapted to the actual business objective. The 0.35 threshold was selected to improve F1, but a production application might instead prioritize recall, precision, cost-sensitive errors, or fairness constraints.

Overall, the Week 1 workflow demonstrates a complete and reproducible machine learning pipeline from problem definition and data preparation through model development, tuning, diagnostics, interpretation, evaluation, and deployment-ready inference.